DATA SCIENCE:

17/9/26

DATA ANALYTICS:

It is the process of collecting, cleaning, Transforming, Analysing & interpreting data to find useful information and support decision making.

STEP - 1

DATA SOURCING

→ collecting Required data

- primary data - data collected directly
- secondary data - Data already collected by someone

→ Sources like, surveys, feedbacks, existing data sets etc.

STEP - 2

DATA pre processing / DATA Transformation:

- cleaning and preparing data before Analysis
- Handling missing or incorrect data
- converting data to suitable format.

STEP: 3

DATA ANALYSIS:

Examine the data to find patterns trends and useful info..

STEP: 4

DATA MODELLING:

DATA modelling involved relationship b/w tables and organizing data properly for Analysis.

DATA

↓

DATA SENSING

↓

DATA Preprocessing / Transformation

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DATA ANALYSIS

↓

DATA MODELLING

↓

DATA ANALYTICS

↓

What happened?

Past & Present

DATA SCIENCE:

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What will happen?

Future prediction

ML models

EXCEL

SYLLABUS:

1) INTRODUCTION
2) BASIC FUNCTIONS / TEXT FUNCTIONS
3) MATH FUNCTIONS
4) Date & TIME functions
5) FINANCIAL FUNCTIONS
6) sorting and filtering functions
7) Data validations.
8) TEXT to columns.
9) conditional formatting
10) Lookups
11) pivot Tables
12) Power Pivot tables
13) power Query.
14) Data visualization
15) AI with EXCEL
16) Dashboard and report generation.

What is Excel:

Microsoft Excel is a powerful spreadsheet tool use to store, organize and visualise data and analyse.

In data Analytics,

It acts as a beginner friendly yet powerful for handling real world datasets.

Why Excel:

-> easy to learn and use
-> handles structured data efficiently
-> has built in functions and formula
-> ~~Power~~ -> visualization tool
-> No coding required.

Structured data = Tabulated data

KEY COMPONENTS of EXCEL:

1) WORKBOOK AND WORKSHEETS:

Workbook -> Excel file (.xlsx)

Worksheet -> alphabets (A, B, C, ...)

2) ROWS AND COLUMNS:

Rows -> NUMBERS (1, 2, 3, 4, ...)

COLUMNS -> alphabets (A, B, C, ...)

3) CELLS:

Intersection of Row and columns:

Eg: (A1)

TYPES of DATA IN EXCEL:

* TEXT -> Names, categories
* NUMBERS -> sales, Quantity
* Dates -> time based analysis
* Boolean -> True (1) | false (0)
* Geographical data -> Latitudes and Longitudes.

IMPORTANT EXCEL FEATURES:

1) DATA EXTRACTION

2) DATA cleaning | Data transformation

* Removes duplicates
* handling the missing values

* Removing irrelevant data and adding Necessary data

2) formules and functions:

* Basic -> sum, average, count
* Logical -> if, and, or, not
* Lookup -> lookup, xlookup, vlookup, hlookup
* Text -> concat, upper, lower

3) Data visualization:

* charts: bar, col, pie ... etc
* conditional formatting
* sparkl lines

4) PIVOT TABLE:

* summarize tables
* Group and filter data
* Generate insights quickly

5) sorting and filtering

6) Data validation:

FUNCTIONS :

set of instruction / set of code that is already pre defined is called function

3 main components of function :

1) function name -> what operation has to be performed
2) Number of inputs -> cell refrences that will give to function
3) Return value. -> The Result after processing input.

EXCEL TEXT FUNCTIONS :

1) LOWER ( ) :

converts all letters into Lower case.

SYNTAX :

= LOWER (Text)

Eg: = LOWER (MANI)

o/p = mani

2) UPPER ( )

converts all letters into UPPER case

SYNTAX:

= UPPER (TEXT)

Eg: = UPPER (gopal)

o/p : GOPAL

3) PROPER ( )

converts first letter of each word into capital and Remaining letter to Lower case.

SYNTAX:

= PROPER (Text)

Eg: = PROPER (mani kannan)

o/p: Mani Kannan.

4) LEN ( )

counts the Number of characters in a text including spaces.

SYNTAX:

= LEN (Text)

Eg: ①

= LEN (mani)

o/p: 4

Eg: ②

= LEN (mani kannan)

o/p: 11

5) LEFT ()

Extracts a specified number of characters in a text, including spaces.

SYNTAX :

LEFT (text, NUM - chars)

Eg: A2 = MANI KANNAN

= LEFT (A2, 3)

o/p : MAN

6) RIGHT () :

Extracts specified Number of characters from Right side.

SYNTAX :

RIGHT (Text, Num - chars)

Eg: A_2 = MANI KANNAN

= RIGHT (A2, 4)

o/p = nan

7) MID ( )

Extracts a specific No of characters starting from given position in the text.

SYNTAX: =MID (text, start_num, Num_chrs)

Eg:

A_2 = MANI KANNAN

=MID (A_2, 6, 3)

O/P = KAN

8) FIRST HALF of NAME using functions

LEFT; Roundup; LEN:

SYNTAX: = LEFT (TEXT, ROUNDUP(LEN(Text)/2,0))

9) SECOND HALF of NAME USING FUNCTIONS

RIGHT, LEN:

Here Roundup is Not Needed because we already used in first half for Rounding off.

SYNTAX for FIRST HALF of NAME:

A_1 = MANI KANNAN

FORMULA: =LEFT (A_1, ROUNDUP (LEN(A1)/2, 0))

O/P = MANI K

SYNTAX FOR SECOND HALF of NAME:

FORMULA: =RIGHT (A1, LEN (A2)/2)

O/P = ANNAN

10) TO FIND FIRST NAME:

To find first name we use two functions =Left and =find.

SYNTAX:

=LEFT (A1, find (" ", A1) - 1)

O/P: MANI

11) TO FIND SECOND NAME:

We use =MID, =FIND, =LEN

FORMULA:

=MID (R2, FIND (" ", A1+1, LEN(A2))

O/P = KANNAN.

12) To get FULL NAME:

(i) USING "=CONCAT" FUNCTION we will combine two different cells with a delimeter.

SYNTAX:

=CONCAT (TEXT1, [Text 2], ...)

Ex:

A_1 = MANI A_2 = KANNAN

FORMULA: CONCAT (A_1, " ", A_2)

o/p = MANI KANNAN

If the string with double quotes is not given the spacing won't be there b/w the name.

(ii) Using "TEXTJOIN" function:

SYNTAX: =TEXTJOIN (delimeter, ignore empty text, text1, ...)

using delimeter the text will be seperated

Eg: =TEXTJOIN (" ", 1, A1, A2)

o/p = MANI KANNAN

(3) FRAMING SENTANCE USING "=TEXT JOIN":

Eg: = TEXT JOIN (" ", 1, "Hi", A1, "GM")

o/p = Hi MANI GM

(4) Substitute ()

It is used to substitute a letter with another letter.

SYNTAX:

SUBSTITUTE (text, old_text, new_text)

Eg: A1 = MANI KANNAN

FORMULA: = SUBSTITUTE (A1, "A", "@")

o/p = M@NI K@NN@N.